Reduced QR factorization

Definition: Let $A \in \mathbb{R}^{m\times n}$ have linearly independent columns. A reduced QR factorization is $A = QR$, where $Q \in \mathbb{R}^{m\times n}$ has orthonormal columns and $R \in \mathbb{R}^{n\times n}$ is invertible and upper triangular. Thus $Q^TQ = I_n$.

Gram–Schmidt produces QR: Every matrix A with linearly independent columns has such a factorization with positive diagonal entries $r_{jj} > 0$. In the Gram–Schmidt convention, $r_{jj} = \|\mathbf{v}_j\|$, where $\mathbf{v}_j$ is column $\mathbf{a}_j$ after subtracting its components along $\mathbf{q}_1, \dots, \mathbf{q}_{j-1}$.

Square case: Every invertible square matrix has $A = QR$ with Q orthogonal and R upper triangular with positive diagonal entries; $Q^TQ = QQ^T = I$.

Signs: Without the positive-diagonal convention, simultaneously negating column j of Q and row j of R leaves QR unchanged.

Rectangular Q and projection

For a reduced factorization with $m > n$, $Q^TQ = I_n$, but generally $QQ^T \neq I_m$. Since $\operatorname{col}(Q) = \operatorname{col}(A)$, for $\mathbf{b} \in \mathbb{R}^m$,

$$QQ^T\mathbf{b} = \sum_{i=1}^{n} (\mathbf{q}_i \cdot \mathbf{b})\mathbf{q}_i = \operatorname{proj}_{\operatorname{col}(Q)}(\mathbf{b}) = \operatorname{proj}_{\operatorname{col}(A)}(\mathbf{b}).$$

Least squares with QR

For $A \in \mathbb{R}^{m\times n}$ with linearly independent columns, $\mathbf{b} \in \mathbb{R}^m$, and a reduced factorization $A = QR$:

Normal equations: $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$ reduce to $R\hat{\mathbf{x}} = Q^T\mathbf{b}$. This upper-triangular system has a unique solution because R is invertible.

Least-squares coefficient and fitted vector: Solve $R\hat{\mathbf{x}} = Q^T\mathbf{b}$ by back-substitution. The fitted vector is

$$A\hat{\mathbf{x}} = QR\hat{\mathbf{x}} = QQ^T\mathbf{b}.$$